

Vamsi Dobbala

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years designing and optimizing enterprise-scale data pipelines, ETL/ELT processes, and cloud-based analytics solutions. Specialized in Snowflake, dbt, and PySpark. Experienced integrating Generative AI capabilities — LLMs, OpenAI APIs, and Model Context Protocol (MCP) — into business workflows. Strong foundation in data modeling, warehousing, data quality, and modern analytics architectures supporting AI-driven solutions.

TECHNICAL SKILLS

Languages	SQL · Python 3.x · PySpark · DAX · PowerShell
Frameworks	Apache Spark · dbt · Delta Lake · Delta Live Tables (DLT) · ETL/ELT · Data Modeling
Cloud & Platforms	Snowflake · Microsoft Fabric · Azure VMs · BigQuery · GCP
BI & Analytics	Power BI · Microsoft Excel · Apache Superset
AI & APIs	OpenAI APIs · Model Context Protocol (MCP) · MCP Server & Connector Development · Claude Skill Creation · GenAI Integration
Tools	Git · GitHub · VS Code · dbt Cloud · Fivetran · AlloyDB · FastAPI

WORK EXPERIENCE

Data Engineer — Bilvantis May 2023 – Present

- Designed layered dbt architectures (Staging → Intermediate → Mart → Metrics) with reusable Jinja/SQL models, cutting development cycle time.
- Built Snowflake data models — fact tables, dimension tables, and reporting datasets — aligned to business requirements.
- Implemented dbt testing frameworks: schema tests, freshness validations, data integrity checks, and custom business rule tests.
- Authored Snowflake stored procedures and automated data validation workflows, ensuring high-quality downstream reporting.
- Optimized complex SQL transformations, reducing query execution times and improving warehouse cost efficiency.
- Delivered curated datasets powering Power BI dashboards; collaborated with business stakeholders on KPI logic definition.
- Managed Git-based source control, PR reviews, CI/CD deployment, and production support under Agile methodologies.

Key Achievements

- Reduced manual KPI validation effort ~90% via automated dbt regression and data quality testing framework.
- Improved cross-system data reliability and stakeholder trust through systematic dbt test coverage.
- Architected scalable Snowflake warehouse supporting enterprise-grade analytics and reporting workloads.

KEY PROJECTS

Claims & Written Member Mapping Snowflake · dbt · Fivetran · SQL · Python · Git

Stack: Source Systems → Fivetran → Snowflake → dbt → Power BI

- Built dbt models across Staging, Intermediate, Mart, and Metrics layers for insurance analytics reporting.
- Managed Fivetran ingestion pipelines; designed fact/dimension models supporting reporting requirements.
- Implemented schema tests, data quality checks, and reconciliation processes to ensure data integrity.
- Optimized Snowflake queries and dbt workflows for improved performance and maintainability.

Limosa M2C – Futuro Data Access Platform BigQuery · GCP · Python · SQL · AlloyDB · FastAPI · Apache Superset

- Designed pipelines migrating payment and transaction data from legacy platforms into GCP (BigQuery).
- Built low-latency transaction lookup datasets enabling efficient customer support workflows.
- Implemented secure role-based access controls; built Apache Superset dashboards for ops and business teams.
- Developed metadata-driven reporting datasets and branded Excel reports for external stakeholders.

DAX Measure Regression Testing & Data Quality Validation Framework Snowflake · dbt · SQL · Python · Power BI · PowerShell

- Built automated regression framework validating business-critical Power BI DAX measures against source-system calculations.
- Developed dynamic expected-value generation in Snowflake via stored procedures and SQL transformations.

- Automated validation with Python + PowerShell — reduced manual effort ~90%.

Claude × Apache Superset – Natural Language BI via MCP

[Anthropic Claude](#) · [MCP](#) · [Apache Superset](#) · [Python](#) · [FastAPI](#)

- Built MCP connector integrating Claude AI with Apache Superset, enabling natural language interaction with dashboards, datasets, and charts without requiring SQL or BI expertise.
- Enabled business stakeholders to query and analyze data directly through conversational AI — eliminating dependency on data analysts for ad-hoc reporting and insight extraction.
- Empowered data analysts to create, configure, and publish Superset charts using natural language instructions via Claude agents, reducing chart-build time significantly.
- Developed Claude skill definitions and MCP tool schemas exposing Superset APIs — covering dataset querying, chart creation, dashboard management, and filter application — as agent-callable actions.

EDUCATION

SRM Institute of Science & Technology

B.Tech – Electronics & Communication Engineering | 2022 | 8.67 CGPA

CERTIFICATIONS

- [SnowPro Associate](#) — Snowflake
- [Microsoft Certified: Power BI Data Analyst Associate \(PL-300\)](#) — Microsoft
- [Snowflake Data Warehouse Essentials](#) — Snowflake
- [Snowflake Collaboration & Marketplace](#) — Snowflake
- [Snowflake Data Applications](#) — Snowflake
- [Anthropic Claude 101](#) — Anthropic